

LEVEL: 2 MEDIUM

Q1



Write a program to find the path with the maximum sum from the top left corner to the bottom right corner of a matrix, moving only right or down.

Sample Input

Q2



Enter number of rows: 3
Enter number of columns: 3
Enter elements

Q3



5 3 2

1 7 1

4 6 8

Q4



Sample Output

Maximum sum path: 5 → 3 → 7 → 6 → 8

Maximum sum: 29

1 / 4 saved

Test Case 1: Enter number of rows: 3 Enter number of columns: 3 Enter elements: 1 2 3 4 5 6 7 8 9

Test Case 2: Enter number of rows: 2 Enter number of columns: 2 Enter elements: 1 2 3 4

Test Case 3: Enter number of rows: 4 Enter number of columns: 4 Enter elements: 1 0 0 0 0 2 0 0 0 0 1 0 0 0 0 4

Your Code

```
1 #include<iostream>
2 #include<vector>
3 using namespace std;
4 int main() {
5     int r, c;
6     cin >> r >> c;
7     vector<vector<int>> dp(r, vector<int>(c))
8     dp[0][0] = a[0][0];
9     for (int i = 1; i < c; i++) {
10         dp
11     }
12 }
```

Output

I

← Back to top

Q4

4 / 4 saved

Test Case 2: Enter a string: Pack my box with five dozen liquor jugs.

Test Case 3: Enter a string: Sphinx of black quartz, judge my vow.

Your Code

```
1 #include<iostream>
2 #include<string>
3 #include<cctype>
4 using namespace std;
5 int main() {
6     string s;
7     cout<<"enter a string ";
8     getline(cin, s);
9
10    bool present[26]={false};
11    for(char c:s){
12        if(isalpha(c)){
13            c=tolower(c);
14            present[c-'a']=true;
15        }
16    }
17    for(int i=0;i<26;i++){
18        if(!present[i]){
19            cout<<"the string is not a pangram";
20            return 0;
21        }
22    }
23    cout<<"the string is pangram";
24    return 0;
25 }
```

Output

enter a string the string is pangram

← Previous

Next →

Test case-1:
Enter number of elements : 6
Enter elements : 7 8 9 10 11 12

Your Code

```
1 #include<iostream>
2 using namespace std;
3 void majority(int a[],int n){
4     for(int i=0;i<n;i++){
5         int count=0;
6         for (int j=i+1;j<n;j++){
7             if(a[i]==a[j])
8                 count ++;
9
10            if(count>n/2)
11                cout<<"majority element"<<a[i];
12            return;
13        }
14    }
15    cout<<"no majority element";
16 }
17 int main(){
18     int n;
19     cin>>n;
20     int a[100];
21     for(int i=0;i<n;i++){
22         cin>>a[i];
23     }
24     majority(a,n);
25     return 0;
26 }
```

Output

no majority element

← Previous

Next →

QUESTIONS

LEVEL 2 — MEDIUM

- Q1
function ✓
- Q2
1DArray ✓
- Q3
string ✓
- Q4
2DArray ✓

4 / 4 saved

Level 2 — Medium gcd (M) — function

✓ Submitted

Write a function `int gcd(int a, int b)` that computes the greatest common divisor of two numbers `a` and `b`.

Sample Input:

Enter two numbers: 56 98

Sample Output:

GCD of 56 and 98 is 14

Your Code

```
1 #include<iostream>
2 using namespace std;
3 int gcd (int a,int b){
4     while (b!=0)
5     {
6         int temp=b;
7         b=a%b;
8         a=temp;
9     }
10    return a;
11 }
12 int main(){
13     int a,b;
14     cout<<"enter the number"<<endl;
15     cin>>a>>b;
16     cout<<a<<endl<<b<<endl;
17     cout<<"the gcd of the number is"<<gcd(a,b);
18     return 0;
19 }
```

Output

```
enter the number
56
98
the gcd of the number is14
```

Next →